



CLOTHING RENTAL

Submitted in partial fulfillment of the requirements of the degree of
Bachelor of Engineering (Information Technology)

By

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Clothing Rental

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Sign and Grade	

Title : Clothing Rental

Project Description :

Clothing Rental is a full-stack web platform designed to simplify the process of renting clothes for various occasions including weddings, parties, casual outings, and photoshoots. The system enables users to browse collections by category, check availability, reserve garments, and manage their rental schedules via a personalized dashboard. It includes secure login, real-time inventory updates, and multiple payment options. Built using the MERN stack (MongoDB, Expr.

Requirement gathering :

Requirements were gathered through surveys and interviews with fashion-conscious individuals, event planners, and college students. Core issues identified included limited access to affordable designer wear, inefficient inventory visibility, and lack of flexible return policies. The solution addresses these challenges by implementing a digital closet with filters, rental duration customization, and easy payment integration. Both renters and inventory managers were considered for a scalable, dual-interface solution.

System Requirements :

Hardware:

- Processor: Intel i3/i5 or AMD Ryzen (dual-core+)
- RAM: 8 GB minimum
- Storage: 1 GB available space
- Internet: Required for OTP & MongoDB Atlas

Software:

- OS: Windows/macOS/Linux
- Python 3.8+
- MongoDB Atlas or Local MongoDB
- Visual Studio Code
- Git

Technologies Used :

Layer	Technology
Frontend	HTML, CSS, Bootstrap, JS
Backend	Python (Flask), Flask-Login
Database	MongoDB
Hosting	Render (Backend)
Analytics	Google Analytics (GA4)

Setup Instructions :

To set up the Microscope Booking System, several components need to be configured properly. First, ensure that Python 3.8 or higher is installed from the official Python website. During installation on Windows, it's important to check the option to add Python to the system PATH. After installing, you can verify it using `python --version` and `pip --version`. It's recommended to create a virtual environment using `python -m venv venv` to manage dependencies separately. This environment can be activated with `venv\Scripts\activate` on Windows or `source venv/bin/activate` on macOS/Linux.

Once Python is set up, install the necessary packages using a `requirements.txt` file, which includes Flask, Flask-Login, Flask-SocketIO, pymongo, dnspython, and python-dotenv. These packages support user authentication, real-time updates, and MongoDB integration.

The system uses MongoDB as its backend database, which can be set up either locally or via MongoDB Atlas. For local setups, MongoDB Community Edition should be installed and started on the default port. For cloud setups, MongoDB Atlas allows you to create a free cluster, set up a user, whitelist your IP, and generate a connection string. This string should be securely stored in a `.env` file along with the Flask `SECRET_KEY` and email credentials if OTP functionality is used.

After all configurations, the application can be launched using `python app.py`, and accessed via `http://localhost:5000`. If deploying to a production server like Render, Gunicorn is used to run the Flask app. With this setup, the Microscope Booking System can run seamlessly, supporting real-time bookings, user authentication, and data persistence via MongoDB.

Backend Setup

1. Navigate to your project folder: `cd microscope-booking-system`
2. Create a virtual environment:
`python -m venv venv`
3. Activate the virtual environment:
`venv\Scripts\activate`
4. Install required Python dependencies:
`pip install -r requirements.txt`
5. Start the Flask development server:
`python app.py`

Frontend Setup

The frontend of the Microscope Booking System is built using standard web technologies — HTML5, CSS3, JavaScript, and Bootstrap — and is rendered through Flask templates.

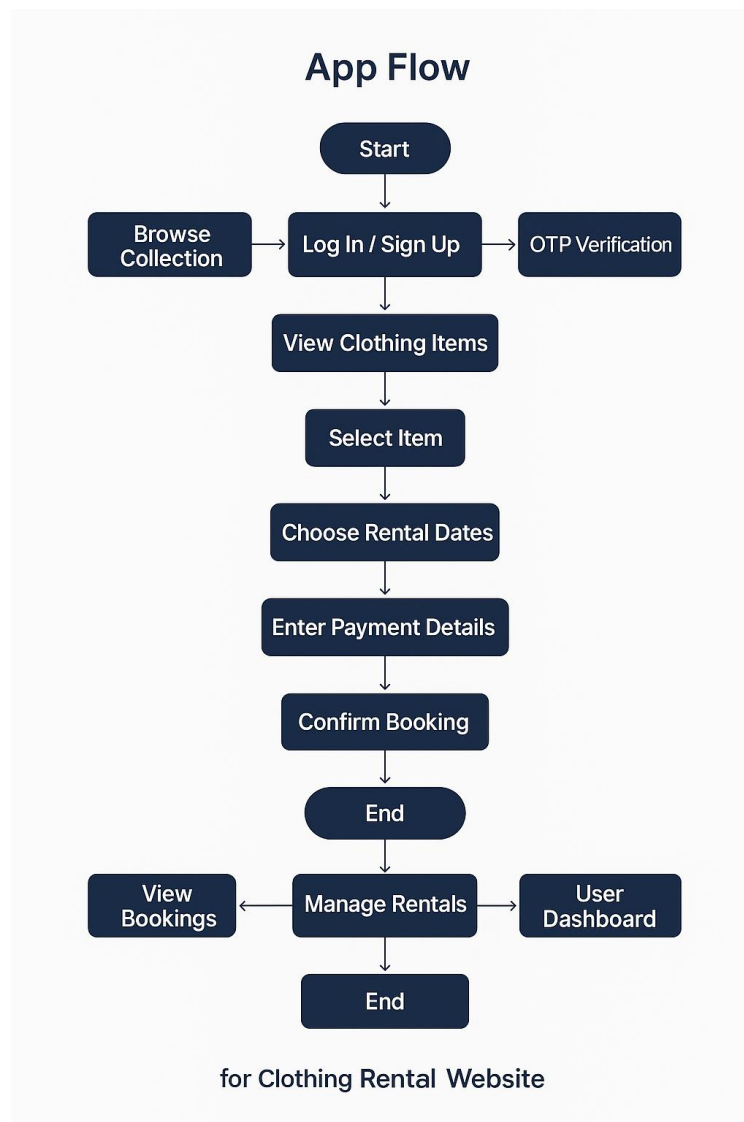
Access the Website

- Once the server starts, visit: **<http://localhost:5000>**

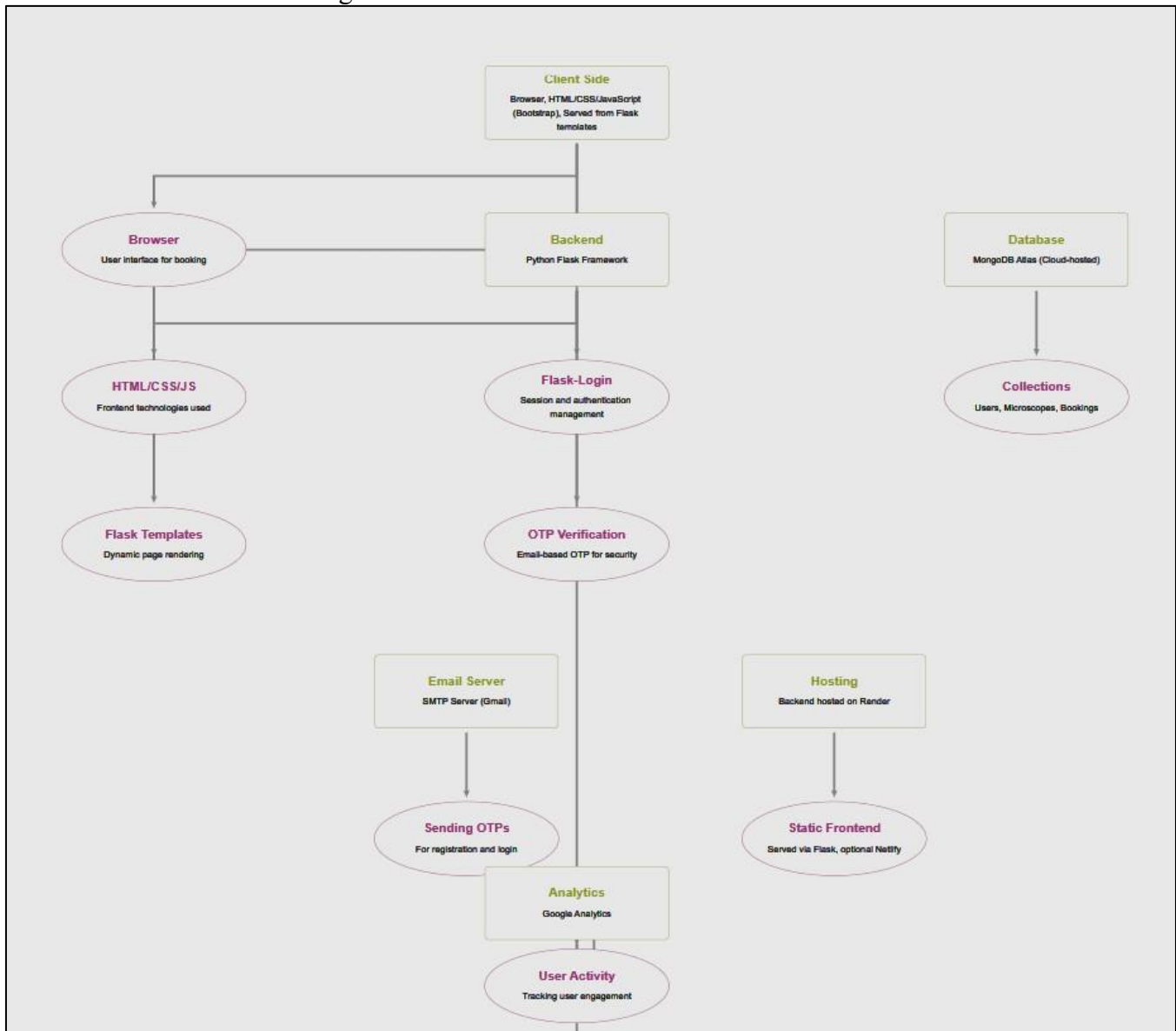
Project Structure:

Architectural Diagrams :

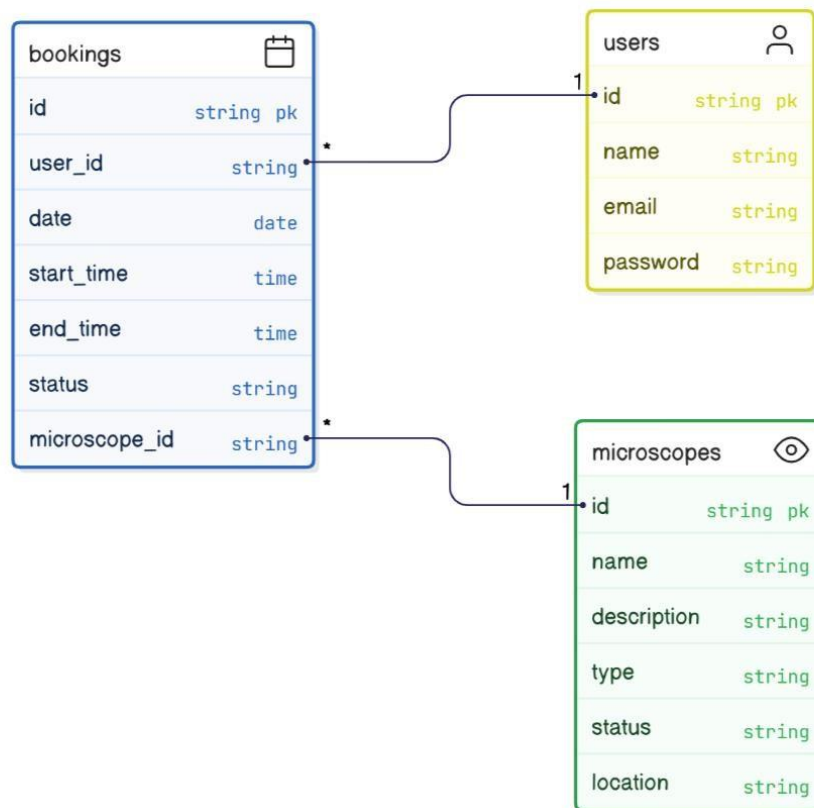
a) Web Flow Diagram -



b) System Architecture
Diagram :



c) UML diagram :



Features Implemented :

- User Authentication (JWT + OTP): Secure sign-up/login with email verification.
- Real-Time Availability: Inventory updates based on active rentals.
- Booking Engine: Choose garment, rental duration, and delivery address.
- Payment Gateway Integration: Rent garments securely with Razorpay.
- Admin Panel: Add/edit garments, manage returns and orders.
- Responsive UI: Tailwind-powered frontend adapts across devices.
- Rental History & Dashboard: Users can view current and past orders.

Screenshots of implementation :

Home Page –

Rent-A-LookHomeBrowse

LoginRegister

Logged out successfully

Rent the Perfect Look

Discover our collection of premium clothing available for rent

[Browse Collection](#)

Featured Items





Elegant Evening Gown
\$50/day

[View Details](#)



Casual Summer Dress
\$25/day

[View Details](#)



Party Jumpsuit
\$35/day

[View Details](#)



Business Suit
\$45/day

[View Details](#)



Casual Jeans
\$20/day

[View Details](#)



Cocktail Dress
\$40/day

[View Details](#)



Formal Blazer
\$55/day

[View Details](#)



Summer T-Shirt
\$15/day

[View Details](#)

localhost:5000/product/680695c765e356533098b1d

Login –

[Rent-A-Look](#) [Home](#) [Browse](#) [Login](#) [Register](#)

Login

Email address

Password

[Login](#)

Don't have an account? [Register here](#)

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Rentals Page –

[Rent-A-Look](#) [Home](#) [Browse](#) [My Rentals](#) [Logout](#)

My Rentals



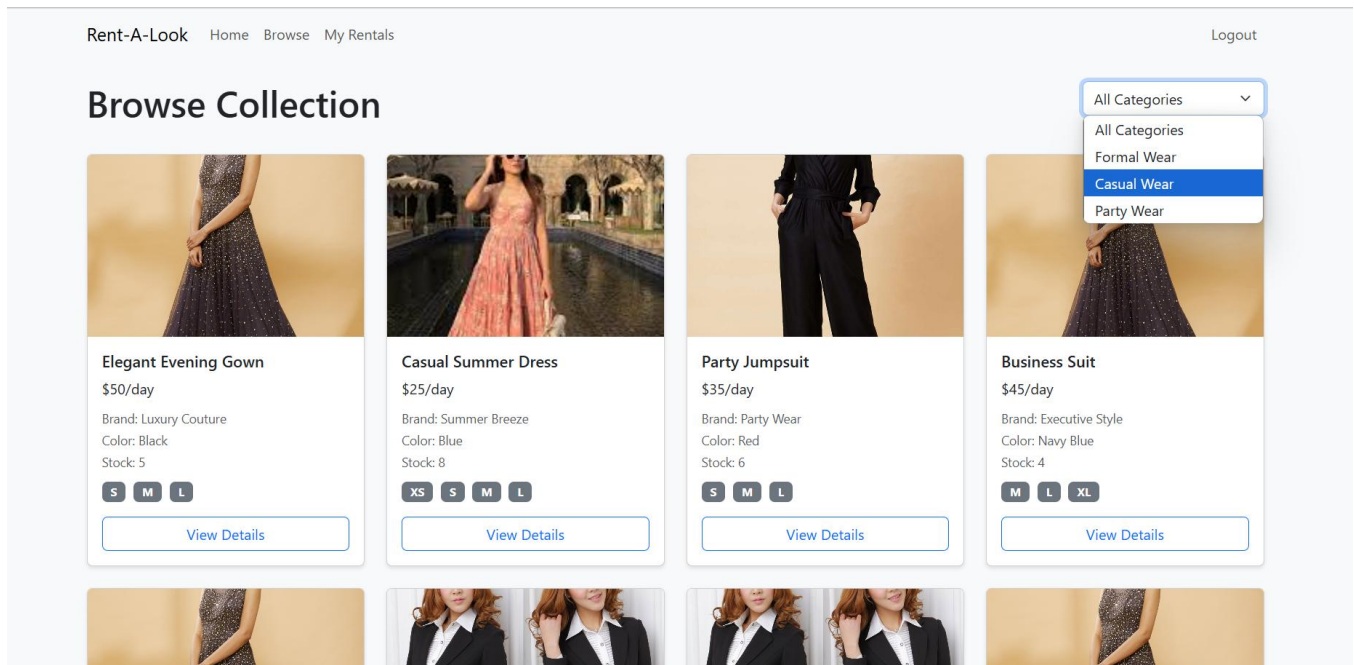
Casual Summer Dress
Rental Date: 2025-04-22
Return Date: 2025-04-22
Status: Returned
XS S M L
Brand: Summer Breeze
Color: Blue



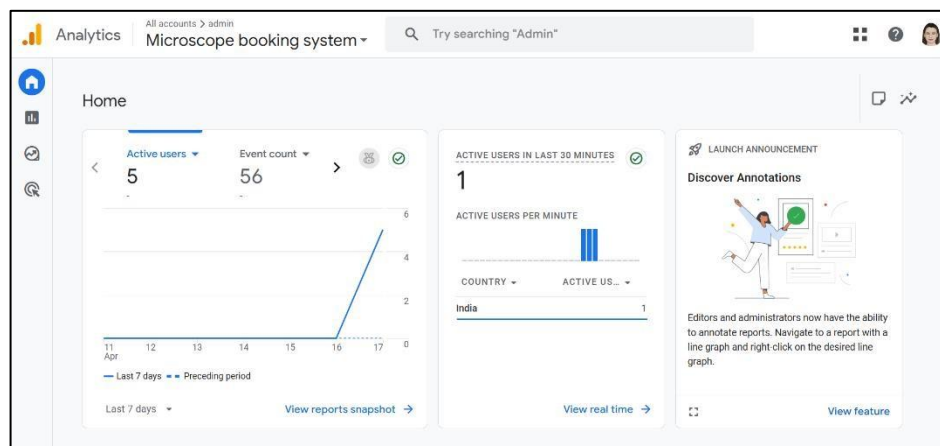
Formal Blazer
Rental Date: 2025-04-22
Return Date: 2025-04-22
Status: Returned
M L XL
Brand: Professional Wear
Color: Charcoal

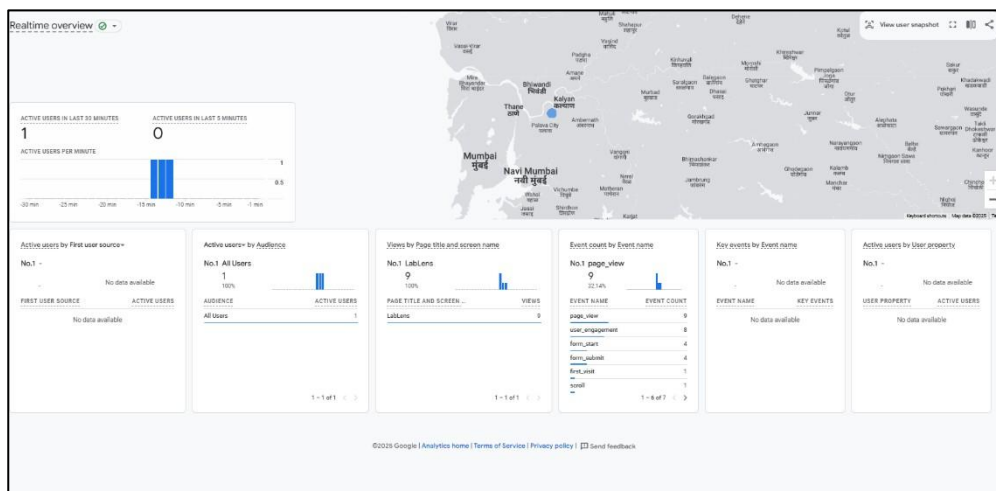
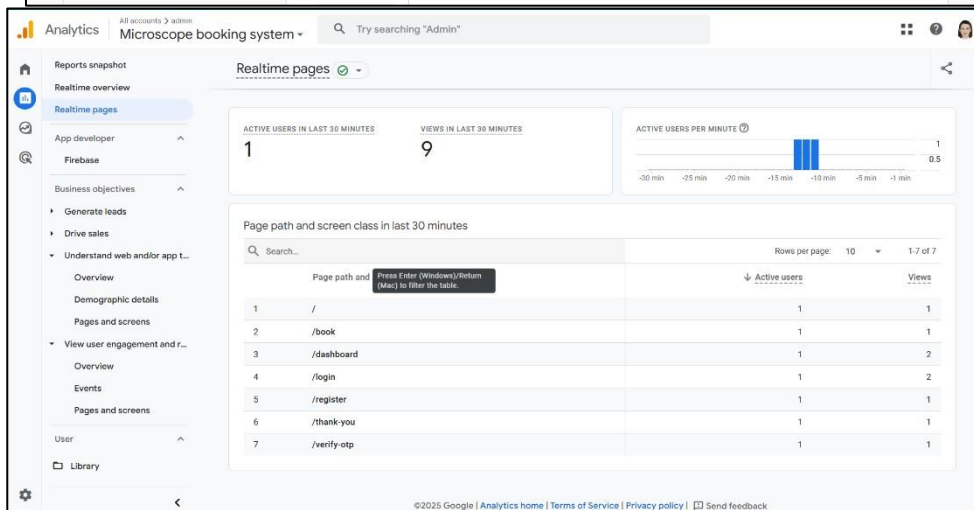
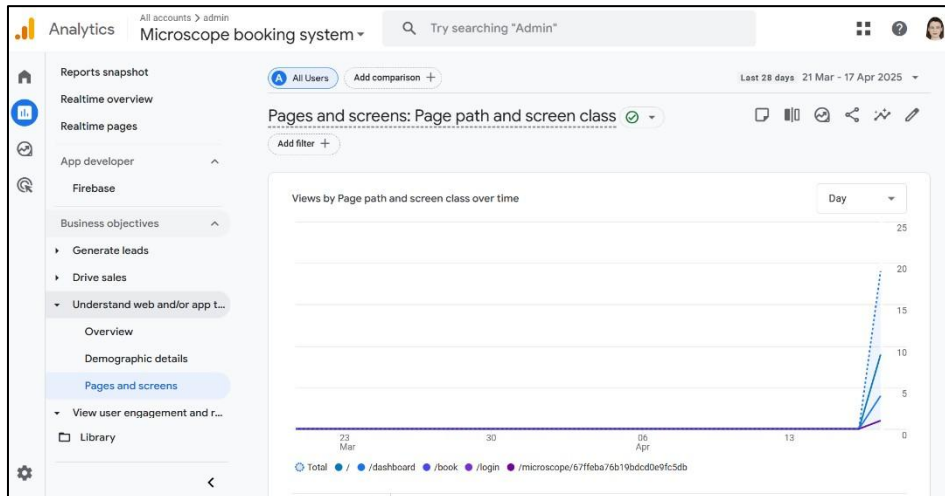


Formal Blazer
Rental Date: 2025-04-22
Return Date: 2025-04-22
Status: Returned
M L XL
Brand: Professional Wear
Color: Charcoal



Google Analytics :





Database (MongoDB) -

users

webx_microscope

- bookings
- microscopes
- user_preferences
- users**

Find Indexes Schema Anti-Patterns Aggregation Search Indexes

Generate queries from natural language in Compass

Filter Type a query: { field: 'value' } Reset Apply Options

INSERT DOCUMENT

```
{
  "_id": ObjectId("67ffede66a13c831eaff7f30"),
  "email": "96sannidhi@gmail.com",
  "name": "sannidhi",
  "password": "pbkdf2:sha256:600000$gE46THMd0uKJU9lg$4c8e58738ba9c7e6ce8f2d4922b989b3...",
  "otp_verified": true
}
```

```
{
  "_id": ObjectId("68024cbe8e3319f42c93dd57"),
  "email": "arnav.sawantbhosale@gmail.com",
  "name": "arnav sawant",
  "password": "pbkdf2:sha256:600000$i1H9yz7WZsyQjauA$ebcbbfd4b09254ee23ebda5e7c5693d3...",
  "otp_verified": true
}
```

Bookings:

Search Namespaces

STORAGE SIZE: 36KB LOGICAL DATA SIZE: 2.29KB TOTAL DOCUMENTS: 12 INDEXES TOTAL SIZE: 180KB

Find Indexes Schema Anti-Patterns Aggregation Search Indexes

Generate queries from natural language in Compass

Filter Type a query: { field: 'value' } Reset Apply Options

INSERT DOCUMENT

QUERY RESULTS: 1-12 OF 12

```
{
  "_id": ObjectId("67ffee286a13c831eaff7f31"),
  "user_id": ObjectId("67ffede66a13c831eaff7f30"),
  "microscope_id": ObjectId("67ffeba76b19bdc08e9fc5db"),
  "date": "2025-04-18",
  "start_time": "16:21",
  "end_time": "19:21",
  "status": "confirmed",
  "created_at": 2025-04-16T23:21:36.781+00:00,
  "updated_at": 2025-04-17T00:27:12.627+00:00
}
```

```
{
  "_id": ObjectId("67fff472090d31bdf8ad4a07"),
  "user_id": ObjectId("67ffede66a13c831eaff7f30"),
  "microscope_id": ObjectId("67ffeba76b19bdc08e9fc5db"),
  "microscope_name": "Electron Microscope",
  "date": "2025-04-18",
  "start_time": "16:13",
  "end_time": "16:48",
  "status": "confirmed",
  "created_at": 2025-04-16T23:21:36.781+00:00,
  "updated_at": 2025-04-17T00:27:12.627+00:00
}
```

Github Link : https://github.com/akkds1022/webx_miniproject.git

Future Scope

- AI-powered outfit recommendations based on occasion and preferences.
- Integration with courier APIs for return pickups and tracking.
- Mobile app version for Android and iOS.
- Loyalty program with credits for frequent renters.
- Try-before-you-rent AR feature for virtual fitting.

Conclusion

Clothing Rental addresses a growing demand for sustainable fashion through a digital rental platform. Using modern web technologies, it offers an intuitive, secure, and scalable solution for both consumers and inventory managers. The integration of user-centric features, responsive design, and a robust backend ensures smooth operations while reducing the cost and waste associated with fast fashion.